

Example 1. Let n be an integer. Prove that $3 \mid n(n^2 + 2)$.

Solution. We divide into cases based on the remainder of n upon division by 3.

Case 1. $n =$ for some integer k

Then: $n(n^2 + 2) =$

which is a multiple of 3.

Case 2. $n =$ for some integer k

Then: $n(n^2 + 2) =$

which is a multiple of 3.

Case 3. $n =$ for some integer k

Then: $n(n^2 + 2) =$

which is a multiple of 3.

□

A. Gcd. We list the positive common divisors of two integers, like 24 and 90.

The integers will have a **greatest common divisor** provided:

The notation is: $\gcd(24, 90) =$

Recall: an integral combination of a, b has form $ax + by$ where x, y are integers.

Observe that $\gcd(24, 90)$ is an integral combination of 24 and 90.

$\gcd(24, 90) =$

This is not a coincidence, as the following section will demonstrate.

B. **Bézout.** We provide an alternative interpretation of the gcd.

Bézout's Identity. Let a and b be integers, not both zero. Then $\gcd(a, b)$ is:

Proof. The set of positive integral combinations of a and b is nonempty because:

By the well-ordering principle, it contains a least element, which we call d .

We write d as an integral combination of a and b .

$d =$

We show that d is a common divisor of a and b .

First, we show that $d \mid a$. Apply the division algorithm.

$a =$

We solve for r as an integral combination of a and b .

$r =$

Note that r cannot be positive, because otherwise:

which contradicts:

Thus $r = 0$, which means $d \mid a$.

An analogous argument shows $d \mid b$, and thus d is a common divisor of a and b .

Next, we show that d is the greatest among common divisors.

Let d' be a common divisor of a and b .

Last class we showed that a common divisor of a and b also divides every:

But d is an:

So follows that $d' \mid d$.

Because d is positive, this implies:

□

A quick consequence of a result or a proof of that result is called a:

Corollaries. Let a and b be integers, not both zero.

- Every common divisor of a and b also:
- Among positive common divisors of a and b , $\gcd(a, b)$ is:

Example 2. Let a and b be integers, not both zero. Let k be a positive integer.

Prove that $\gcd(ka, kb) = k \gcd(a, b)$.

Proof. For ease of notation, write $d = \gcd(a, b)$.

Then the goal is to show:

First, we show that kd is a positive common divisor of ka and kb .

kd is positive because:

Last class, we showed that, if $x \mid w$ and $y \mid z$, then:

Since $k \mid k$ and $d \mid a$ it follows that:

Since $k \mid k$ and $d \mid b$ it follows that:

So, kd is a common divisor of ka and kb .

Next, we show that kd is an integral combination of ka and kb .

Write d as an integral combination of a and b .

Next multiply by k .

Thus, kd is an integral combination of ka and kb .

But the only pos. common divisor that is also an integral combination is the gcd.

Thus $kd = \gcd(ka, kb)$.

□

C. **Relatively Prime.** We calculate:

$$\gcd(18, 25) =$$

When $\gcd(a, b) = 1$, we say a and b are **relatively prime**, and we write:

Let a and b be integers, not both zero. Then $a \perp b$ if and only if:

A statement of the form “ P if and only if Q ” means both:

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Essentially, it asserts that P and Q are logically equivalent statements.

We often contract “if and only if” to:

So, in our case there are actually two proofs we need to write.

Proof. $\implies$: We assume:

and prove:

Because $\gcd(a, b) = 1$, Bézout’s identity implies:

$\impliedby$: We assume:

and prove:

Because 1 is an integral combination of a and b , it follows that:

Hence, $1 = \gcd(a, b)$.

□